

R.J. THOMPSON

Senior Software Developer · C++ / C# · Engines, Tooling & Real-Time Systems

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Senior software developer shipping production software since 1996, across AAA games, enterprise systems and medical imaging. Shipped titles at BioWare (Dragon Age), Activision (Call of Duty), Sega/Relic (Company of Heroes 3) and Electronic Arts, including five years on the Frostbite/FrostEd engine and editor tooling. Specialist in the hard-to-staff work: low-level threading and networking, engine and editor frameworks, and performance optimization on large shipping codebases. Experienced team lead and mentor, comfortable owning a system end to end under shipping deadlines.

TECHNICAL SKILLS

Languages	C++ (17), C#, Python, Lua, JavaScript, SQL
Engines	Frostbite / FrostEd, Essence Editor, Unity, Unreal, Havok
Frameworks	.NET, WPF, XAML, MVVM, Win32 API, Electron
Specialties	Threading & debugging, low-level networking (Sockets, TCP/IP), real-time systems, graphics & animation optimization, engine/editor tooling
Platforms	Windows, Switch, PlayStation, Xbox, iOS, Android, embedded
Tools	Perforce, Git, Jenkins, Visual Studio, Azure DevOps

SELECTED ACHIEVEMENTS

- Optimized **Company of Heroes 3** within the live-ops team for a measured **7% performance gain**, and fixed process issues that lifted team velocity by 10%.
- Shipped the dialog and narrative system for **Dragon Age 4** on the Frostbite engine.
- Ported low-level networking and real-time voice chat to Nintendo Switch for **Operation Tango**.
- Accelerated **Skylanders** character animation asset-bundle loading by **200%**; designed the dynamic character scaling system that kept rigs Havok-compatible.
- Built real-time video overlay for **Star Wars BB-8** holding a steady 30fps on low-spec mobile.
- Delivered the **Call of Duty: Ghosts** companion app under a hard three-month deadline.

PROFESSIONAL EXPERIENCE

Utherverse — Metaverse Platform

2024 – 2025

Assistant to the Director of Development · six-month contract

- Ran daily operations of a live virtual-world platform, coordinating cross-functional teams to keep the user experience stable.
- Owned departmental planning with the Director, plus staffing, onboarding and executive reporting.

BioWare (Electronic Arts) — Dragon Age

2023 – 2024

Senior Software Developer · seven-month contract

- Shipped the dialog and narrative system for Dragon Age 4.
 - Mentored developers on the FrostEd cinematic and narrative systems, drawing on five prior years on EA's cinematics team.
- Tech: C++, Frostbite, FrostEd · Platforms: Switch, Android, iOS

Activision — Call of Duty

2022 – 2023

Senior Software Developer · six-month contract

- Shipped a new live-ops system for Call of Duty.
 - Ported low-level networking and real-time voice chat for Operation Tango to Switch; resolved Photon networking defects.
 - Debugged difficult low-level threading and networking issues on Disney Mirrorverse.
 - Third-party integration specialist (Vivox, Photon, OpenCV); architected improved interop COM bridges for Android and iOS.
- Tech: C++, Unity · Platforms: Switch, Android, iOS

Sega / Relic Entertainment — Company of Heroes 3

2021 – 2022

Senior Software Developer

- Optimized WPF, Relic's C++ port of WPF, inside the Essence Editor 6 engine; delivered up to 7% performance improvement.
- Built the front-end UI, loadout system, and HUD for terrain and unit indicators.
- Architected an improved UI API and authored the documentation for the proprietary engine.
- Decompiled the .NET framework to keep the C++ WPF framework behaviourally aligned with WPF.
- Wrote build tooling for Perforce pipelines and hot-reload; fixed process issues that improved team velocity by 10%.

Tech: C++17, Essence Editor 6, WPF, XAML

Electronic Arts

Oct 2016 – Sept 2021

Software Developer

- Core contributor to the FrostEd editor across the Editor Frameworks, Cinematics, Schematics and Lighting workflow teams.
- Fundamental engine tech: WPF, threading, docking, performance and memory profiling.
- Architected MVVM upgrades to a legacy visual-scripting diagram library.
- Built cinematic features used daily by artists — grouping, video and keyframe controls, playback/rewind, preview-window HUD; helped architect the keyframe editor.
- Wrote the crash control and recovery application; added Azure telemetry and dashboard support.

Tech: C++, C#, WPF/XAML, Frostbite / Ignite

Finger Food Studios

Apr 2014 – Oct 2016

Development Team Lead

- Led the studio's transition to Unity, improving project efficiency and cross-team collaboration.
- Architected and shipped the Skylanders Imaginators mobile app for iOS and Android in Unity.
- Developed a proprietary non-uniform animation scaling system keeping rigs compatible with the console Havok engine.
- Optimized a proprietary GPU-to-CPU shared texture memory system in C#; integrated console comms via async Winsock and low-level TCP/IP.
- Achieved 30fps on low-end mobile for Star Wars BB-8 real-time video overlay.

Projects: Call of Duty, Star Wars BB-8, Skylanders, MaxPlay · Tech: C#, C++, Unity, JavaScript

Industrial Alliance Insurance

Aug 2013 – Feb 2014

Software Engineer

- Architected the Unification business project in .NET using WPF, WCF and MVVM.
- Wrote all client-side WCF connections; profiled web-service performance and improved scalability and error handling.

Microsoft — BigPark

Apr 2012 – Aug 2013

Development Team Lead

- Built complex real-time UI systems in Lua for NFL, ESPN, UFC and Bonnaroo on Xbox.
- Implemented a system visualizing live NFL player positions from helmet telemetry and GPS.

Tech: Lua, C++, XAML, Unity, Kinect

Novadaq Technologies

2012 – Apr 2013

Software Developer

- Lead developer and architect of the Cinevaq 2.0 (SPY ELITE) product line — fluorescence imaging for surgical use.

Tech: C# with C++ video libraries

FinancialCAD

2010 – Apr 2012

Software Developer

- Lead architect of the Silverlight/XNA presentation layer and of the Fair Value insight reporting project.
- Optimized N-tier scalability by writing a faster .NET XML parser.

Neoteric Technology

Jul 2004 – Jun 2010

Software Developer

- Built UI libraries from the ground up with the Win32 API in C++; designed the N-tier backend.
- Designed the blood-management process and architecture; delivered Auto Fate Web in ASP.NET 2.0.
- Led a team of four developers with hiring and training responsibility; owned UML design docs, test plans and automated suites.

Max Integration

Jan 2004 – Jul 2004

Software Developer / Project Lead

- Designer and project lead of Look Out, a Win32 CRM flagship product; grew the team to six developers.
- Designed the N-tier backend on SQL 2000 and converted the product to mobile via ASP.NET.

RedHawk Gaming Software

2001 – 2003

Software Developer

- Designed gaming subsystems and low-level device drivers for barcode readers; maintained legacy code and supported QA.

Infowave Technologies

1997 – 2001

Quality Assurance Lead

- Managed a QA team of five; maintained client and server features, glib compression and optimization; developed automated test suites.

Electronic Arts

1996 – 1997

Quality Assurance

- Built test suites for NBA 97 (Saturn), Need for Speed 1–2 (PC), Warcraft II, NHL 98 (PSX & Saturn), PGA Tour 97–98 (PSX).

EDUCATION AND TRAINING

BCIT — .NET High Tech Professional Software Systems Development (2003–2004); Advanced C++ (1999–2001)**WSA Consulting** — UML (2000); OO Techniques and Large-Scale Application Design; Development Testing and Automation Tools (1998–2001)